Tokenis: Cryptographic Pseudonymization and Idea Protection
Using EC-ElGamal for Privacy-Preserving AI Systems

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February 4, 2026

Abstract

As AI systems increasingly rely on private and proprietary data, conventional data anonymization and tokenization techniques have proven insufficient to prevent information leakage during retrieval-augmented generation (RAG) and autonomous AI agent execution. Surface-level identifier masking leaves the deeper semantic structure of domain knowledge—decision rules, scoring formulas, and operational procedures—fully exposed to reconstruction through iterative querying. This paper presents **Tokenis**, a cryptographic pseudonymization framework that combines Named Entity Recognition (NER)–based data selection with elliptic-curve ElGamal (EC-ElGamal) encryption to protect both personal data and high-value domain ideas. Unlike existing approaches that focus solely on personally identifiable information (PII), Tokenis introduces an *idea protection protocol* that prevents the reconstruction and replication of proprietary rules, strategies, and algorithms while still enabling AI-driven reasoning over the protected knowledge. We present a formal security model grounded in the Decisional Diffie-Hellman (DDH) assumption, prove three core security properties (PII confidentiality, idea reconstruction resistance, and adaptive query safety), and demonstrate end-to-end integration with the TorusDB RAG platform. Tokenis serves as a foundational privacy layer for encrypted RAG systems and autonomous AI agencies, enabling controlled utilisation of sensitive domain knowledge without exposing it to the model.

1 Introduction

Large Language Models (LLMs) have become a core component of modern AI systems. Their effectiveness in enterprise and research settings, however, depends critically on access to high-quality domain knowledge: internal policies, proprietary strategies, and curated datasets that represent years of accumulated expertise. Deploying these models therefore raises a fundamental tension between *utility* and *confidentiality*—the system must reason over sensitive knowledge without ever possessing it in recoverable form.

Traditional anonymization and tokenization methods are inadequate in this context for two distinct reasons. First, they primarily protect surface-level identifiers rather than semantic knowledge; replacing a name or an account number does nothing to shield the conditional logic or algorithmic structure embedded in the surrounding text. Second, and more critically, they

fail to prevent the reconstruction of sensitive logic through AI-generated outputs: an adversary who issues sufficiently many adaptive queries can piece together protected rules even when all identifiers have been masked.

This paper argues that privacy-preserving AI systems require protection not only for personal data, but also for *ideas*—high-level rules, procedures, and decision logic embedded in documents. We introduce Tokenis, a cryptographic framework that enforces a strict separation between AI reasoning and knowledge possession. The system operates as a preprocessing layer that intercepts raw documents, detects both PII and idea-level assets, encrypts them under EC-ElGamal, and delivers only policy-controlled, auditable access to the downstream AI pipeline.

The remainder of the paper is organised as follows. Section 2 formalises the problem setting and threat model. Section 3 states the design goals. Sections 4–7 detail the NER-based pseudonymisation pipeline and the idea protection protocol. Section 8 exploits the additive homomorphism of EC-ElGamal for policy evaluation without decryption. Section 9 presents the formal security model and proof sketches for three core theorems. Section 11 compares Tokenis with the state of the art. Section 12 describes end-to-end integration with TorusDB RAG. Finally, Section 13 concludes.

2 Problem Statement

2.1 Limitations of Existing Pseudonymisation Techniques

Most pseudonymisation systems focus exclusively on replacing identifiers such as names, email addresses, or account numbers. Formally, given a document D, these systems detect a set of entity spans $\{m_i\}$ and replace each with a synthetic token $\{\tau_i\}$ while leaving the rest of the document unchanged. While this transformation is sufficient for achieving regulatory compliance under frameworks such as GDPR Article 4(5), it does not protect the structural knowledge that surrounds those identifiers.

Consider a document that encodes a proprietary pricing strategy: "If the client tier is Platinum and the quarterly volume exceeds 500 units, apply a 17% discount with an additional 3% loyalty bonus." A standard pseudonymiser will mask "Platinum" or "500 units" at best, yet the conditional structure, the threshold values, and the discount formula remain fully visible. An adversary observing AI outputs generated from this document can reconstruct the complete pricing logic through a small number of targeted queries.

The elements left unprotected by conventional approaches include internal decision rules and conditional logic, proprietary pricing or scoring strategies, and algorithmic procedures embedded as natural language. In AI-driven systems these elements constitute the most strategically valuable part of the knowledge base, and yet they are precisely what existing pseudonymisation leaves exposed.

2.2 Threat Model

Adaptive semantic adversary. We consider a deterministic or probabilistic adversary $\mathcal{A}$ that interacts with the Tokenis-protected system in an adaptive, multi-round fashion. The defining characteristic of $\mathcal{A}$ is its ability to perform *semantic reconstruction attacks*: rather than targeting individual identifiers, the adversary aims to infer underlying decision logic, strategic

rules, or proprietary procedures by correlating AI outputs across rounds. Each query issued by $\mathcal{A}$ may depend arbitrarily on the full transcript of all previous interactions, capturing realistic attack patterns observed in retrieval-augmented generation (RAG) systems and autonomous agent deployments.

Observable system surface. The adversary is assumed to have full visibility into all AI-generated outputs returned by the system, including intermediate reasoning text when exposed by the application layer. In addition, $\mathcal{A}$ may access encrypted knowledge base entries and any retrieval metadata or transcripts explicitly made available through the system interface. This models a powerful but practical attacker who does not need to breach cryptographic primitives, but instead exploits legitimate system behavior and observable outputs.

Trust and exclusion assumptions. We explicitly do not trust the AI model with respect to confidentiality. The model is treated as a potentially leaky component that may reveal sensitive information if it is present in recoverable form. Consequently, Tokenis does not rely on prompt engineering, alignment, or behavioral constraints to enforce privacy. Conversely, the adversary is assumed not to possess any decryption key managed by Tokenis; cryptographic key management is delegated to a trusted authority outside the threat model. This separation allows the analysis to focus on semantic leakage and reconstruction resistance rather than key compromise.

3 Design Goals

The design of Tokenis is guided by the observation that privacy-preserving AI systems must address threats that arise not only from direct data exposure, but also from semantic leakage through inference and repeated interaction. Accordingly, the framework is built around four orthogonal design goals. Together, these goals define the minimal requirements for a cryptographically sound privacy layer that can be deployed in retrieval-augmented generation (RAG) systems and autonomous AI agents without sacrificing operational utility.

G1 — Cryptographic pseudonymisation. Tokenis adopts a strictly cryptographic notion of pseudonymisation in which all sensitive values are protected by semantically secure encryption rather than reversible string substitution. In contrast to conventional tokenisation schemes that rely on lookup tables or deterministic mappings, Tokenis enforces that every entity value emitted from the preprocessing pipeline is represented as an EC-ElGamal ciphertext. This construction ensures indistinguishability under chosen-plaintext attacks (IND-CPA) and prevents linkage attacks across documents or sessions. Crucially, no plaintext identifier or idea-bearing fragment is allowed to propagate downstream into the AI pipeline. By eliminating reliance on heuristic masking or syntactic obfuscation, cryptographic pseudonymisation aligns the system's privacy guarantees with formal security definitions rather than empirical assumptions about model behavior.

G2 — Idea-level protection. While cryptographic protection of identifiers is necessary, it is not sufficient in AI-driven systems where the most valuable assets often take the form of abstract

logic rather than concrete data fields. Tokenis therefore elevates proprietary logic to a first-class protected asset. Decision rules, scoring formulas, thresholds, and procedural knowledge embedded in natural language are explicitly identified and treated as *idea spans*. Rather than masking identifiers within such text, the framework intercepts the idea spans themselves and subjects them to cryptographic protection. This design prevents adversaries from reconstructing business logic or strategic knowledge through repeated or adaptive querying, even in scenarios where no individual PII element is ever revealed. By extending pseudonymisation from data to ideas, Tokenis directly addresses a class of semantic leakage attacks that remain unmitigated by existing approaches.

G3 — AI-compatible reasoning. Strong confidentiality guarantees are only meaningful if the resulting system remains usable. Tokenis is therefore designed to preserve the operational utility of AI systems while enforcing strict separation between reasoning and knowledge possession. Downstream models are able to perform inference and decision-making tasks over the protected knowledge base without access to plaintext entities or idea text. This capability is achieved through a combination of policy-controlled reveal levels and the exploitation of the additive homomorphism of EC-ElGamal for limited policy evaluation. As a result, the AI system can condition its outputs on protected knowledge and enforce access policies without ever possessing sensitive content in a recoverable form. This goal distinguishes Tokenis from approaches that rely on trusted models or post-hoc output filtering.

G4 — Auditability and governance. In enterprise and regulated environments, privacy guarantees must be verifiable and enforceable over time. Tokenis therefore embeds auditability as a core system property rather than an afterthought. Every access to protected knowledge results in the generation of a cryptographic receipt that records which encrypted idea tokens were accessed, the reveal level applied, and the policy decision that authorized the access. These receipts form a tamper-evident audit trail that supports post-hoc compliance verification, governance enforcement, and forensic analysis in the event of suspected misuse or leakage. By coupling cryptographic protection with explicit accountability mechanisms, Tokenis enables privacy-preserving AI deployments that remain operationally transparent and governable.

4 System Overview

Tokenis is architected as a cryptographic preprocessing layer positioned between raw knowledge sources and the downstream AI pipeline. Its primary function is to enforce a strict separation between *knowledge possession* and *AI reasoning*. Rather than relying on the model to behave securely, Tokenis ensures that sensitive content is never presented to the model in plaintext form. A document entering the system is transformed through a sequence of four stages, each corresponding to a distinct security and functionality boundary.

Stage 1 — Joint entity and idea detection. The first stage identifies the semantic units that require protection. Conventional named entity recognition (NER) is applied to detect personally identifiable information (PII) such as names, identifiers, and account references. In parallel, Tokenis applies an Idea-Keyphrase Recognition (IKR) module that targets higher-level semantic constructs, including decision rules, conditional logic, scoring formulas, thresholds,

and procedural descriptions. These detected regions, referred to as *entity spans* and *idea spans*, define the minimal set of content that must be cryptographically protected to prevent both direct data leakage and semantic reconstruction attacks.

Stage 2 — Cryptographic encryption and tokenisation. In the second stage, all detected entity and idea spans are encrypted using EC-ElGamal under a system-managed public key. Each plaintext span is replaced in situ by a synthetic token that serves as a stable reference within the document while revealing no semantic information. The encrypted ciphertexts are stored separately in a side-channel ciphertext store, indexed by their corresponding tokens. The output of this stage is a token-annotated document paired with an encrypted ciphertext repository, which together form the *protected knowledge base*. At this point, no recoverable sensitive information remains in the text accessible to the AI pipeline.

Stage 3 — Policy-aware retrieval. When the AI system issues a query, retrieval is mediated by a policy engine that determines how each referenced token may be handled. Policies encode access constraints, reveal levels, and contextual conditions under which partial information may be derived. Crucially, this decision process occurs before any content is exposed to the model. Depending on the applicable policy, a token may remain fully opaque, be resolved to a derived or abstract representation, or participate in limited homomorphic evaluation without decryption. This stage ensures that access control is enforced at the semantic level rather than through post-hoc filtering of model outputs.

Stage 4 — AI generation over protected knowledge. In the final stage, the large language model performs reasoning and text generation over the policy-filtered view of the knowledge base. The model operates exclusively on tokenised text and authorised derived values, never receiving plaintext entities or idea spans. As a result, the AI system can produce contextually meaningful outputs while remaining cryptographically incapable of reconstructing the underlying protected knowledge. This design ensures that confidentiality guarantees hold independently of the model's internal behavior or alignment properties.

Raw Document

↓

NER + Idea Detection (IKR)

↓

EC-ElGamal Encryption + Tokenisation

↓

Protected Knowledge Base

↓

Policy-Aware Retrieval

↓

LLM Generation

This four-stage pipeline formalises a clear trust boundary: sensitive knowledge is cryptographically protected before it ever interacts with the AI model, and all subsequent access is mediated by explicit, auditable policies. By construction, Tokenis eliminates reliance on trusted model behavior and replaces it with enforceable cryptographic guarantees, making the system suitable for deployment in adversarial and high-value knowledge environments.

5 Idea Assets and Functional Abstraction

A central contribution of Tokenis is the explicit protection of *idea-level assets*. In contrast to personally identifiable information (PII), which admits relatively clear syntactic definitions, the notion of an "idea" is often criticised as being vague or ill-defined. This section resolves that ambiguity by formalising idea assets not as textual fragments, but as computational objects defined by their functional role in decision-making. This abstraction provides a precise boundary between data literals and protected ideas, and forms the foundation for the security guarantees proved in Section 9.

5.1 Idea Assets as Computational Objects

We emphasise that Tokenis does not attempt to protect "meaning" in a linguistic or philosophical sense. Instead, it protects behavioural logic: information whose disclosure enables reproduction of decision procedures, scoring rules, or operational policies.

Definition 1 (Idea Asset (Computational Form))**.** *An* idea asset *is a computational object extracted from text that induces a decision-relevant mapping between inputs and outputs. Formally, an idea asset ι is defined as a triple*

$$\iota := (\mathcal{X}, \mathcal{Y}, f),$$

where $\mathcal{X}$ is an input domain, $\mathcal{Y}$ is an output domain, and $f : \mathcal{X} \to \mathcal{Y}$ is a (possibly partial or stochastic) function implicitly specified by the underlying text.

Under this definition, isolated facts or constants are not idea assets unless they participate in a functional dependency. The defining property of an idea asset is not its surface form, but its role in constraining or determining system behaviour.

5.2 Idea Compilation from Text

Let Σ denote the space of textual spans extracted from documents. Tokenis applies an *idea compilation* step that maps text into functional abstractions.

Definition 2 (Idea Compilation)**.** *An idea compilation is a (possibly randomized) mapping*

$$\mathsf{Comp} : \Sigma \to \mathcal{F},$$

where $\mathcal{F}$ is a space of evaluable functions. For a textual span $\sigma \in \Sigma$, $\mathsf{Comp}(\sigma) = f_\sigma$ denotes the function induced by σ.

The compilation step discards the original textual representation after processing. All subsequent reasoning, policy evaluation, and security analysis are conducted with respect to the induced function f_σ rather than the original text σ.

5.3 Functional Equivalence Classes

The abstraction above allows us to define a precise notion of when two textual descriptions represent the same protected idea.

Definition 3 (Functional Equivalence Class)**.** *Let $f : \mathcal{X} \to \mathcal{Y}$ be a function induced by an idea asset. The* functional equivalence class *of f is defined as*

$$[f] := \{\sigma \in \Sigma \mid \mathsf{Comp}(\sigma) = f\}.$$

All elements of $[f]$ are textually distinct but computationally indistinguishable under black-box access to the induced function.

Intuitively, two textual descriptions belong to the same equivalence class if they induce identical observable behaviour for all admissible inputs. Tokenis protects idea assets at the granularity of these equivalence classes, rather than at the level of individual textual realisations.

5.4 Boundary Between Data and Ideas

The functional abstraction provides a clear boundary between data literals and idea assets.

Example (Numeric literals versus idea assets). A numeric literal such as "17%" or a phrase such as "17% discount" does not, in isolation, define an idea asset. However, the statement

"If the client tier is Platinum, apply a 17% discount"

induces a function

$$f(x) := \begin{cases} 0.83 \cdot p(x) & \text{if } x \in \mathsf{Platinum}, \\ p(x) & \text{otherwise}, \end{cases}$$

mapping client attributes to pricing outcomes, and is therefore classified as an idea asset. Distinct textual formulations that induce the same f belong to the same functional equivalence class and are protected identically by Tokenis.

5.5 Role of the IKR Module

The Idea-Keyphrase Recognition (IKR) module is responsible for identifying textual spans whose functional contribution warrants protection. Crucially, IKR is not required to perform perfect semantic understanding.

Soundness requirement. We require only that the IKR module be *sound*: every span it classifies as an idea asset must correspond to a functional dependency. False negatives may reduce coverage but do not weaken the security guarantees for detected assets.

Functional detection criterion. Formally, a textual span σ is flagged as an idea asset candidate if there exists an input $x \in \mathcal{X}$ such that

$$\mathsf{Comp}(\sigma)(x) \neq \mathsf{Comp}(\sigma \setminus \text{span})(x),$$

i.e., removing the span would alter the induced function. This criterion abstracts away from specific implementation choices and allows IKR to be realised via rule-based patterns, dependency parsing, or learned models, without affecting the subsequent security analysis.

5.6 Implications for Security Analysis

By formalising idea assets as functional equivalence classes, Tokenis shifts the security objective from protecting textual secrecy to preventing functional reconstruction. The adversary may observe the behaviour of f_σ under policy-mediated access, but cannot determine which element of $[f_\sigma]$ was originally present. This abstraction underlies the information-theoretic reconstruction bounds proved in Section 9 and clarifies precisely what it means for an idea to remain protected.

6 NER-Based Cryptographic Pseudonymisation

6.1 Entity Detection

Let T denote an input text of length n tokens. A Named Entity Recognition function

$$\mathcal{N}(T) \;=\; \big\{\,(t_i,\; s_i,\; e_i)\,\big\}_{i=1}^{k}$$

returns a set of k detected entities, where t_i denotes the entity type, and the span $[s_i, e_i)$ identifies the character-level extent of entity i within T. The protected entity set is defined as

$$\mathcal{S}_{\text{PII}} \;=\; \{\,\texttt{PERSON},\ \texttt{EMAIL},\ \texttt{ID},\ \texttt{ACCOUNT},\ \texttt{PHONE},\ \texttt{ADDRESS},\ \ldots\,\}.$$

Any detected entity whose type belongs to $\mathcal{S}_{\text{PII}}$ is forwarded to the encryption stage. Entities that do not match any protected type are left in place, preserving the grammatical structure of the document for downstream reasoning.

6.2 EC-ElGamal Encryption

Let $\mathbb{G} = \langle G \rangle$ be an elliptic curve group of prime order q, and let $H_{\text{curve}} : \{0,1\}^* \to \mathbb{G}$ be a hash-to-curve function modelled as a random oracle. The key pair is generated as follows: the secret key is a uniformly random value $x \xleftarrow{\$} \mathbb{Z}_q$, and the corresponding public key is $P = xG$.

Given a plaintext entity value $m_i \in \{0,1\}^*$, encryption proceeds in three steps. First, the entity is mapped onto the curve:

$$M_i \;=\; H_{\text{curve}}(m_i) \;\in\; \mathbb{G}.$$

Second, a fresh randomness value is sampled:

$$r_i \;\xleftarrow{\$}\; \mathbb{Z}_q.$$

Third, the ciphertext pair is computed:

$$C_i \;=\; \big(\,r_i\,G,\;\; M_i + r_i\,P\,\big) \;=\; (C_i^{(1)},\; C_i^{(2)}).$$

The ciphertext C_i is stored in the side-channel ciphertext store, and the plaintext m_i is replaced

in T by a deterministic token τ_i that serves as a pointer to C_i. From this point forward the AI pipeline operates exclusively over tokens; it never observes any m_i or any decryption key.

6.3 Correctness

Decryption recovers the curve point via $M_i = C_i^{(2)} - x\, C_i^{(1)}$, since

$$C_i^{(2)} - x\, C_i^{(1)} \;=\; \big(M_i + r_i\, x\, G\big) - x\, r_i\, G \;=\; M_i.$$

The original string m_i is then recovered by inverting H_{curve} over a pre-image table maintained by the trusted key-management authority.

7 Idea Protection Protocol

7.1 Definition of Idea Assets

The core contribution of Tokenis is the elevation of *ideas* to first-class protected assets. To avoid ambiguity, Tokenis does not define ideas at the level of textual meaning, but at the level of computational behaviour.

Definition 4 (Idea Asset)**.** *An* Idea Asset *is a computational object induced by a textual span that defines or constrains a decision-relevant mapping between inputs and outputs. Formally, an idea asset is a tuple*

$$\iota := (\mathcal{X}, \mathcal{Y}, f),$$

where $\mathcal{X}$ is an input domain, $\mathcal{Y}$ is an output domain, and $f : \mathcal{X} \to \mathcal{Y}$ is a (possibly partial or stochastic) function implicitly specified by the text.

A textual span σ is associated with an idea asset ι if and only if removing σ from the document would alter the induced function f. Under this definition, isolated facts or numeric literals are not idea assets unless they participate in a functional dependency.

Unlike PII, idea assets are not atomic identifiers. They represent *structural knowledge*: self-contained units of logic whose value derives from the relationships among conditions, thresholds, and actions. Masking a single identifier within an idea asset does not neutralise it, as the adversary may reconstruct the functional logic from the remaining structure. Tokenis therefore treats each idea asset as an indivisible unit for the purpose of protection.

7.2 Idea Detection via Functional Sensitivity

Tokenis applies an Idea-Keyphrase Recognition (IKR) process after PII entities have been extracted. The role of IKR is not to interpret semantics, but to identify textual spans whose presence affects the induced computational behaviour.

Formally, a textual span $\sigma \subseteq T$ is flagged as an `IDEA-Span` if there exists an input $x \in \mathcal{X}$ such that

$$\mathsf{Comp}(T)(x) \;\neq\; \mathsf{Comp}(T \setminus \sigma)(x),$$

where Comp denotes the idea compilation function mapping text to a functional representation. Each detected span is assigned a unique idea identifier ι_j for $j = 1, \ldots, \ell$, where ℓ is the number of idea assets in the document.

This criterion abstracts away from specific implementation choices. In practice, IKR may be realised via rule-based patterns (e.g., conditional constructs, thresholds, procedural verbs), syntactic analysis, or learned models, but the security guarantees of Tokenis do not depend on any specific detection technique.

7.3 Classification of Idea Assets

Not every idea asset requires the same protection semantics. Tokenis classifies each detected idea asset according to the data owner's policy into one of the following categories.

Reversible Idea Assets. A reversible idea asset is one whose textual representation may be recovered under explicit policy authorisation. Such assets are encrypted under EC-ElGamal in the same manner as PII entities. The ciphertext is stored in a side-channel store and may be decrypted only by a trusted key-management authority. During AI operation, only policy-controlled derived information (e.g., the existence or high-level purpose of a rule) is released to the model; the full text is never exposed.

Non-Reversible Idea Assets. A non-reversible idea asset is protected against textual reconstruction by construction. After detection, the textual span inducing the idea is compiled into a functional representation f suitable for policy evaluation or scoring, and the original text is permanently discarded. No textual representation of the idea exists in the system state after preprocessing.

As a consequence, an adversary interacting with the system may observe the *behaviour* of the idea through policy-mediated outputs, but cannot recover the textual rule or algorithm that produced it. This property underlies the information-theoretic reconstruction bounds established in Section 9.

8 Homomorphic Properties and Policy Evaluation

EC-ElGamal inherits the additive homomorphic property of the underlying group operation. Given two ciphertexts $C_1 = (r_1 G,\ M_1 + r_1 P)$ and $C_2 = (r_2 G,\ M_2 + r_2 P)$, their component-wise sum yields

$$C_1 + C_2 \ = \ \big((r_1 + r_2)\, G,\ \ (M_1 + M_2) + (r_1 + r_2)\, P \big),$$

which is a valid encryption of $M_1 + M_2$ under randomness $r_1 + r_2$.

Tokenis exploits this property to perform *encrypted policy evaluation*. A policy rule of the form "grant access if the sum of the client's tier scores exceeds a threshold θ" can be evaluated entirely in ciphertext space: the individual tier-score ciphertexts are summed homomorphically, and the result is compared against an encryption of θ using an encrypted equality or range check. The policy engine never sees any plaintext score; it issues an access decision based solely on the relationship between ciphertexts.

This mechanism is essential for Goal G3 (AI-compatible reasoning): it allows the system to enforce fine-grained access controls without decrypting the underlying data, thereby maintaining the separation between the AI pipeline and the protected knowledge base.

9 Formal Security Model and Proof

This section formalises the security guarantees of Tokenis. We first state the cryptographic assumptions on which our proofs rest, then define three security properties and provide proof sketches for each.

9.1 Cryptographic Assumptions

Assumption 1 (Decisional Diffie-Hellman (DDH))**.** *Let $\mathbb{G} = \langle G \rangle$ be an elliptic curve group of prime order q. For all probabilistic polynomial-time distinguishers $\mathcal{D}$,*

$$\Big|\, \Pr\big[\mathcal{D}(G,\ aG,\ bG,\ abG) = 1\big] - \Pr\big[\mathcal{D}(G,\ aG,\ bG,\ cG) = 1\big] \,\Big| \ \leq\ \varepsilon_{\mathrm{DDH}}(n),$$

where $a, b \xleftarrow{\$} \mathbb{Z}_q$ and $c \xleftarrow{\$} \mathbb{Z}_q$ are independent, and $\varepsilon_{\mathrm{DDH}}(n)$ is negligible in the security parameter n.

Assumption 2 (Random Oracle)**.** *The hash-to-curve function $H_{\mathrm{curve}} : \{0,1\}^* \to \mathbb{G}$ behaves as a truly random function.*

9.2 Security Property 1: PII Confidentiality

We show that encrypted PII values are indistinguishable from random group elements under chosen-plaintext attack.

Theorem 1 (PII Confidentiality)**.** *Let $\mathcal{A}$ be a probabilistic polynomial-time adversary playing the standard IND-CPA game against EC-ElGamal on $(\mathbb{G}, G, q)$. Under Assumption 1 and the Random Oracle model (Assumption 2),*

$$\mathrm{Adv}^{\text{IND-CPA}}_{\text{EC-ElGamal}}(\mathcal{A}) \ \leq\ \varepsilon_{\mathrm{DDH}}(n).$$

Proof. We prove the claim by a reduction from the Decisional Diffie–Hellman (DDH) problem. Assume for contradiction that there exists a probabilistic polynomial-time adversary $\mathcal{A}$ that achieves a non-negligible advantage $\varepsilon(n)$ in the IND-CPA game against EC-ElGamal on $(\mathbb{G}, G, q)$.

We construct a DDH distinguisher $\mathcal{D}$ that uses $\mathcal{A}$ as a subroutine. The distinguisher $\mathcal{D}$ is given a tuple

$$(G,\ A,\ B,\ Z) \in \mathbb{G}^4,$$

where $A = aG$, $B = bG$, and Z is either abG or a uniformly random group element, and must decide which is the case.

Simulation of the IND-CPA game. The distinguisher $\mathcal{D}$ sets the EC-ElGamal public key to

$$P := A = aG.$$

Since $\mathcal{D}$ does not know the discrete logarithm a, it cannot decrypt ciphertexts, but this is consistent with the IND-CPA experiment, where the adversary is never given the secret key.

When $\mathcal{A}$ submits two challenge plaintexts $M_0, M_1 \in \mathbb{G}$, $\mathcal{D}$ samples a random challenge bit $b^\star \xleftarrow{\$} \{0,1\}$ and constructs the challenge ciphertext as

$$C^\star := (C_1^\star, C_2^\star) = (B,\ M_{b^\star} + Z).$$

Correctness of the simulation. If $Z = abG$, then

$$C^\star = (bG,\ M_{b^\star} + abG),$$

which is distributed identically to a valid EC-ElGamal encryption of $M_{b^\star}$ under public key $P = aG$ with randomness b:

$$\mathsf{Enc}_P(M_{b^\star}; b) = (bG,\ M_{b^\star} + bP).$$

Hence, in this case, the view of $\mathcal{A}$ is identical to that in the real IND-CPA experiment.

If Z is uniformly random in $\mathbb{G}$, then $C_2^\star = M_{b^\star} + Z$ is also uniformly random and independent of $b^\star$, since group addition is a bijection. Consequently, the entire ciphertext $C^\star$ is information-theoretically independent of the challenge bit, and $\mathcal{A}$ can do no better than random guessing:

$$\Pr[b' = b^\star] = \tfrac{1}{2}.$$

Advantage preservation. After receiving the challenge ciphertext, $\mathcal{A}$ outputs a guess $b' \in \{0,1\}$. The distinguisher $\mathcal{D}$ outputs 1 if $b' = b^\star$, and 0 otherwise. Therefore,

$$\Pr[\mathcal{D}(G, A, B, Z) = 1 \mid Z = abG] = \tfrac{1}{2} + \varepsilon(n),$$

while

$$\Pr[\mathcal{D}(G, A, B, Z) = 1 \mid Z \xleftarrow{\$} \mathbb{G}] = \tfrac{1}{2}.$$

It follows that $\mathcal{D}$ distinguishes DDH tuples from random tuples with advantage exactly $\varepsilon(n)$. Since $\mathcal{D}$ runs in probabilistic polynomial time, the existence of $\mathcal{A}$ with non-negligible advantage would contradict Assumption 1. Therefore, for all probabilistic polynomial-time adversaries $\mathcal{A}$,

$$\mathrm{Adv}^{\text{IND-CPA}}_{\text{EC-ElGamal}}(\mathcal{A}) \le \varepsilon_{\text{DDH}}(n),$$

which completes the proof. □

Theorem 1 guarantees that an adversary who intercepts the ciphertext store learns nothing about the underlying PII values beyond their length, which is consistent with standard semantic security definitions.

9.3 Security Property 2: Idea Non-Reconstructability

We now address the central security claim of Tokenis: that protected idea assets cannot be textually reconstructed from system outputs.

Definition 5 (Idea Reconstruction Attack)**.** *An* idea reconstruction attack *is a strategy* $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ *playing the following game against Tokenis. In phase 1,* $\mathcal{A}_1$ *selects an idea asset* σ^* *and submits a polynomial number of adaptive queries* $q_1, q_2, \ldots, q_{\text{poly}(n)}$ *to the AI system, receiving*

outputs $o_1, o_2, \ldots$. In phase 2, $\mathcal{A}_2$ outputs a candidate text $\hat{\sigma}$. The adversary wins if $\hat{\sigma}$ is semantically equivalent to σ^* (i.e., encodes the same decision logic or algorithm).

Theorem 2 (Idea Reconstruction Resistance)**.** *Let σ^* be a non-reversible idea asset processed by Tokenis. Then for any probabilistic polynomial-time adversary $\mathcal{A}$,*

$$\Pr\big[\mathcal{A} \textit{ wins the Idea Reconstruction Attack}\big] \;\leq\; \varepsilon_{\text{info}}(n),$$

where $\varepsilon_{\text{info}}(n)$ is negligible, provided that the functional compilation of σ^* does not leak its textual structure beyond the observable input-output behaviour.

Proof. We give an information-theoretic proof under an explicit "black-box only" leakage condition.

Setup and leakage condition. Let Σ denote the space of possible idea assets (textual rules, procedures, formulas). Tokenis applies a (possibly randomized) compilation procedure

$$\mathsf{Comp} : \Sigma \to \mathcal{F}$$

that maps an idea asset $\sigma \in \Sigma$ to a *functional abstraction* $f_\sigma \in \mathcal{F}$ (e.g., an executable predicate, a scoring function, or a rule-evaluation oracle). After preprocessing, the system discards the textual representation and retains only f_σ (plus encrypted tokens and audit receipts).

We formalise the statement "the compilation of $\sigma^\star$ does not leak its textual structure beyond observable input–output behaviour" as the following leakage condition: the entire interactive view of any adversary $\mathcal{A}$ interacting with the Tokenis-protected system depends on σ *only* through oracle access to f_σ mediated by policy. Concretely, there exists an oracle $\mathcal{O}_{f_\sigma}$ such that, for every PPT $\mathcal{A}$, the distribution of $\mathcal{A}$'s transcript in the real system is identical to that of $\mathcal{A}$ interacting with $\mathcal{O}_{f_\sigma}$:

$$\mathsf{View}_{\mathcal{A}}^{\mathsf{real}}(\sigma) \;\equiv\; \mathsf{View}_{\mathcal{A}}^{\mathsf{oracle}}(\mathcal{O}_{f_\sigma}).$$

(Here $\equiv$ denotes equality of distributions.)

Equivalence classes induced by black-box behaviour. Fix an idea asset $\sigma^\star \in \Sigma$ and let $f^\star := f_{\sigma^\star}$. Define the behavioural equivalence class of $\sigma^\star$ as

$$\mathcal{E}(\sigma^\star) \;:=\; \big\{\sigma \in \Sigma \;:\; f_\sigma = f^\star\big\}.$$

All $\sigma \in \mathcal{E}(\sigma^\star)$ induce *exactly* the same oracle $\mathcal{O}_{f^\star}$ and hence (by the leakage condition) induce the same adversarial view distribution.

Reconstruction game. Consider the following reconstruction experiment parameterised by the class $\mathcal{E}(\sigma^\star)$.

1. The challenger samples $S \xleftarrow{\$} \mathcal{E}(\sigma^\star)$ uniformly at random and sets $f \leftarrow f_S$.
2. $\mathcal{A}$ is given access to the oracle $\mathcal{O}_f$ (i.e., policy-mediated black-box access) and outputs a guess $\widehat{S}$.
3. $\mathcal{A}$ *wins* if $\widehat{S} = S$.

Let T denote the full interaction transcript between $\mathcal{A}$ and $\mathcal{O}_f$, including all adaptive queries and responses.

Key indistinguishability claim. For any fixed transcript value t in the support of T and any two assets $\sigma_0, \sigma_1 \in \mathcal{E}(\sigma^\star)$, we have

$$\Pr[T = t \mid S = \sigma_0] \;=\; \Pr[T = t \mid S = \sigma_1],$$

because $f_{\sigma_0} = f_{\sigma_1} = f^\star$ implies the oracle behaviour is identical, hence the entire interactive distribution is identical. Therefore, for all $\sigma \in \mathcal{E}(\sigma^\star)$,

$$\Pr[T = t \mid S = \sigma] \;=\; \Pr[T = t \mid S \in \mathcal{E}(\sigma^\star)].$$

Equivalently, T is independent of S:

$$T \;\perp\!\!\!\perp\; S.$$

Posterior remains uniform. Since S is uniform on $\mathcal{E}(\sigma^\star)$ and T is independent of S, Bayes' rule yields, for every $\sigma \in \mathcal{E}(\sigma^\star)$ and every transcript value t,

$$\Pr[S = \sigma \mid T = t] \;=\; \Pr[S = \sigma] \;=\; \frac{1}{|\mathcal{E}(\sigma^\star)|}.$$

Bounding the reconstruction probability. Let $\widehat{S} = \mathcal{A}(T)$ be the adversary's output. Conditioned on any transcript $T = t$, the best possible strategy succeeds with probability at most the maximum posterior mass:

$$\Pr[\widehat{S} = S \mid T = t] \;\le\; \max_{\sigma \in \mathcal{E}(\sigma^\star)} \Pr[S = \sigma \mid T = t] \;=\; \frac{1}{|\mathcal{E}(\sigma^\star)|}.$$

Taking expectation over T gives

$$\Pr[\widehat{S} = S] \;=\; \mathbb{E}_{t \leftarrow T}\big[\Pr[\widehat{S} = S \mid T = t]\big] \;\le\; \frac{1}{|\mathcal{E}(\sigma^\star)|}.$$

Negligibility. By assumption, the compilation is many-to-one with exponentially large preimages under black-box semantics: there exists a function $\kappa(n)$, growing at least linearly in the security parameter, such that

$$|\mathcal{E}(\sigma^\star)| \;\ge\; 2^{\kappa(n)}.$$

Hence,

$$\Pr\big[\mathcal{A} \text{ wins the Idea Reconstruction Attack}\big] \;\le\; 2^{-\kappa(n)} \;=\; \varepsilon_{\text{info}}(n),$$

which is negligible in n. This proves the theorem. □

9.4 Security Property 3: Adaptive Query Safety (Tightened)

Reveal levels and per-event leakage. Tokenis assigns each policy-authorised disclosure a *reveal level* $r \in \mathcal{R}$ (e.g., opaque, derived, partial, full). Each reveal level induces an information-leakage bound $\Delta(r)$ measured in bits. Intuitively, $\Delta(r)$ upper bounds the maximum additional

information about an idea asset that can be conveyed by a single disclosure at reveal level r, even when the adversary is fully adaptive and conditions on the entire past transcript.

Definition 6 (Idea Exposure Budget)**.** *For each idea asset ι_j, the policy engine maintains an* exposure budget $B_j \in \mathbb{N}$. *Each query that triggers a reveal of information related to ι_j consumes at least one unit of B_j. When $B_j = 0$, no further information related to ι_j is released.*

Definition 7 (Per-reveal leakage function)**.** *Fix an idea asset ι_j with plaintext representation σ_j, modeled as a random variable S_j. For each reveal level $r \in \mathcal{R}$, $\Delta(r)$ is an upper bound such that for every round t and every adaptive adversary,*

$$I(S_j;\, R_t^{(j)} \mid \mathsf{Hist}_{t-1}) \le \Delta(r_t) \quad \textit{whenever the reveal level for } \iota_j \textit{ at round } t \textit{ is } r_t = r,$$

where $R_t^{(j)}$ is the reveal payload *for ι_j at round t (defined precisely below), and Hist_{t-1} is the entire interaction history prior to round t.*

Transcript decomposition. To make the data-processing argument explicit, we decompose each round's observable transcript into a policy component and a model-output component. Let T_t denote all information revealed to the adversary in round t. We write

$$T_t \;:=\; (P_t,\, Y_t),$$

where P_t is the *policy transcript* (retrieval metadata, receipts, token identifiers, reveal level decisions, etc.) and Y_t is the *model output* returned by the LLM. The full transcript is

$$T \;:=\; (T_1, \ldots, T_Q) \;=\; (P_1, Y_1, \ldots, P_Q, Y_Q),$$

for $Q \le \mathrm{poly}(n)$.

For a fixed asset ι_j, define the round-t reveal payload $R_t^{(j)}$ as the component of (P_t, Y_t) attributable to the policy's authorised disclosure for ι_j.[1] If no disclosure regarding ι_j occurs in round t, we set $R_t^{(j)} = \perp$.

Theorem 3 (Adaptive Query Safety (Level-Tight))**.** *Let $\mathcal{A}$ be a PPT adversary issuing at most $Q = \mathrm{poly}(n)$ adaptive queries against a Tokenis-protected system. Fix an idea asset ι_j with plaintext representation σ_j, modeled as a random variable S_j. Let $r_t \in \mathcal{R}$ be the reveal level applied to ι_j in round t (with $r_t = \mathsf{opaque}$ when no disclosure occurs), and let B_j be the exposure budget for ι_j.*

Assume the per-reveal leakage function $\Delta(\cdot)$ of Definition 7 holds for all rounds, and the exposure budget enforces a hard cap on the number of non-opaque disclosures:

$$\big|\{t \in [Q] : r_t \neq \mathsf{opaque}\}\big| \;\le\; B_j.$$

Then the total information leakage about ι_j satisfies

$$I(S_j; T) \;\le\; \sum_{t=1}^{Q} \Delta(r_t).$$

[1] Formally, one may define $R_t^{(j)} := \phi_j(P_t, Y_t)$ for a deterministic projection ϕ_j that extracts the asset-j-specific disclosure content (including any derived values) from the total transcript. This only strengthens the bound by the data processing inequality.

In particular, if $\sum_{t=1}^{Q} \Delta(r_t) < H_\infty(S_j)$*, then for any reconstruction attempt* $\widehat{S}_j = g(T)$ *output by* $\mathcal{A}$*,*

$$\Pr[\widehat{S}_j = S_j] \;\le\; 2^{-(H_\infty(S_j) - \sum_{t=1}^{Q} \Delta(r_t))},$$

and exact reconstruction of σ_j *is information-theoretically infeasible.*

Proof. We first upper bound the mutual information between S_j and the full transcript T by summing per-round leakage contributions, and then derive a guessing-probability bound using min-entropy.

Step 1: Reducing leakage to reveal payloads. Since each $R_t^{(j)}$ is a (deterministic) function of the round transcript $T_t = (P_t, Y_t)$, the full sequence

$$R^{(j)} := (R_1^{(j)}, \ldots, R_Q^{(j)})$$

is a deterministic function of T. By the data processing inequality,

$$I(S_j; T) \;\le\; I(S_j; R^{(j)}). \tag{1}$$

Step 2: Chain rule with adaptivity. Apply the chain rule for mutual information:

$$I(S_j; R^{(j)}) \;=\; \sum_{t=1}^{Q} I\big(S_j;\, R_t^{(j)} \mid R_1^{(j)}, \ldots, R_{t-1}^{(j)}\big).$$

Let Hist_{t-1} denote the complete interaction history prior to round t, i.e.,

$$\mathsf{Hist}_{t-1} := (Q_1, P_1, Y_1, \ldots, Q_{t-1}, P_{t-1}, Y_{t-1}).$$

Because $(R_1^{(j)}, \ldots, R_{t-1}^{(j)})$ is a deterministic function of Hist_{t-1}, conditioning on the full history can only reduce mutual information, hence

$$I\big(S_j;\, R_t^{(j)} \mid R_1^{(j)}, \ldots, R_{t-1}^{(j)}\big) \;\le\; I(S_j;\, R_t^{(j)} \mid \mathsf{Hist}_{t-1}). \tag{2}$$

Step 3: Applying the per-level leakage bound. By Definition 7, if the reveal level for ι_j in round t is r_t, then

$$I(S_j;\, R_t^{(j)} \mid \mathsf{Hist}_{t-1}) \;\le\; \Delta(r_t).$$

Combining with (1) and (2) yields

$$I(S_j; T) \;\le\; I(S_j; R^{(j)}) \;\le\; \sum_{t=1}^{Q} I(S_j;\, R_t^{(j)} \mid \mathsf{Hist}_{t-1}) \;\le\; \sum_{t=1}^{Q} \Delta(r_t),$$

proving the first claim.

Step 4: From mutual information to guessing probability via min-entropy. Let $\widehat{S}_j = g(T)$ be any (possibly randomized) estimator output by the adversary. Since $\widehat{S}_j$ is a

function of T, we have again by data processing

$$I(S_j; \widehat{S}_j) \;\le\; I(S_j; T) \;\le\; \sum_{t=1}^{Q} \Delta(r_t).$$

We now bound the success probability $\Pr[\widehat{S}_j = S_j]$ in terms of min-entropy. Recall that the (unconditional) guessing probability of S_j is

$$p_{\text{guess}}(S_j) \;:=\; \max_{s \in \mathcal{S}_j} \Pr[S_j = s] \;=\; 2^{-H_\infty(S_j)}.$$

Conditioned on transcript $T = t$, the optimal guessing probability is

$$p_{\text{guess}}(S_j \mid T = t) \;:=\; \max_{s \in \mathcal{S}_j} \Pr[S_j = s \mid T = t] \;=\; 2^{-H_\infty(S_j \mid T = t)}.$$

Therefore the adversary's success probability is bounded by the conditional guessing probability averaged over T:

$$\Pr[\widehat{S}_j = S_j] \;\le\; \mathbb{E}_{t \leftarrow T}\big[p_{\text{guess}}(S_j \mid T = t)\big].$$

To relate $p_{\text{guess}}(S_j \mid T)$ to leakage, observe that any interactive transcript T that reveals at most $\Lambda := \sum_{t=1}^{Q} \Delta(r_t)$ bits of information can increase the optimal guessing probability by at most a factor 2^{Λ}.[2] Equivalently,

$$H_\infty(S_j \mid T) \;\ge\; H_\infty(S_j) - \Lambda,$$

which implies

$$p_{\text{guess}}(S_j \mid T) \;=\; 2^{-H_\infty(S_j \mid T)} \;\le\; 2^{\Lambda} \cdot 2^{-H_\infty(S_j)} \;=\; 2^{-(H_\infty(S_j) - \Lambda)}.$$

Thus,

$$\Pr[\widehat{S}_j = S_j] \;\le\; 2^{-(H_\infty(S_j) - \Lambda)} \;=\; 2^{-(H_\infty(S_j) - \sum_{t=1}^{Q} \Delta(r_t))}.$$

If $\sum_{t=1}^{Q} \Delta(r_t) < H_\infty(S_j)$, then the above bound is strictly less than 1 and decays exponentially in the remaining entropy gap, implying that exact reconstruction of σ_j is information-theoretically infeasible even under fully adaptive querying. □

Theorem 3 formalises the intuition that bounded, policy-controlled reveals are insufficient for reconstruction, even under the strongest adaptive adversary model. The policy designer sets B_j and the reveal levels to ensure that the product $B_j \cdot \Delta_{\max}$ remains below the entropy threshold for each asset.

Corollary 1 (Budget-to-Leakage Upper Bound)**.** *In the setting of Theorem 3, fix an idea asset ι_j with exposure budget B_j and let $r_t \in \mathcal{R}$ be the reveal level applied to ι_j at round t (with $r_t = \mathsf{opaque}$ when no disclosure occurs). Define the non-opaque index set*

$$\mathcal{I}_j \;:=\; \{t \in [Q] : r_t \neq \mathsf{opaque}\}, \qquad |\mathcal{I}_j| \le B_j.$$

[2] This is a standard min-entropy leakage bound: revealing at most Λ bits (in the sense of an information bound) can increase guessing probability by at most 2^{Λ}. One can view this as a consequence of the fact that a transcript with "Λ bits of information" can shrink the effective candidate set by at most a factor 2^{Λ}. The resulting bound is commonly stated as $H_\infty(S_j \mid T) \ge H_\infty(S_j) - \Lambda$ and hence $p_{\text{guess}}(S_j \mid T) \le 2^{\Lambda} p_{\text{guess}}(S_j)$.

Then the total leakage admits the budget-aware bound

$$I(S_j;T) \;\le\; \sum_{t\in\mathcal{I}_j} \Delta(r_t).$$

Moreover, if we define the worst-case B_j-event leakage

$$\Delta_\star^{(B_j)} \;:=\; \max_{r_1,\dots,r_{B_j}\in\mathcal{R}} \sum_{k=1}^{B_j} \Delta(r_k),$$

then

$$I(S_j;T) \;\le\; \Delta_\star^{(B_j)}.$$

In particular, if $\Delta(r)$ is monotone in a total order of reveal levels and $\Delta_{\max} := \max_{r\neq \mathsf{opaque}} \Delta(r)$, then

$$I(S_j;T) \;\le\; B_j \cdot \Delta_{\max}.$$

Proof. The first claim follows immediately from Theorem 3 by observing that $\Delta(\mathsf{opaque}) = 0$ (since $R_t^{(j)} = \perp$ carries no asset-specific disclosure), hence

$$\sum_{t=1}^{Q} \Delta(r_t) \;=\; \sum_{t\in\mathcal{I}_j} \Delta(r_t) \;+\; \sum_{t\notin\mathcal{I}_j} \Delta(\mathsf{opaque}) \;=\; \sum_{t\in\mathcal{I}_j} \Delta(r_t).$$

Therefore,

$$I(S_j;T) \;\le\; \sum_{t=1}^{Q} \Delta(r_t) \;=\; \sum_{t\in\mathcal{I}_j} \Delta(r_t).$$

For the second bound, note that the sum $\sum_{t\in\mathcal{I}_j} \Delta(r_t)$ contains at most B_j non-zero terms because $|\mathcal{I}_j| \le B_j$. Let the (at most) B_j non-opaque reveal levels that occur be listed as $(\rho_1,\dots,\rho_m)$ where $m := |\mathcal{I}_j| \le B_j$ and each $\rho_k \in \mathcal{R}\setminus\{\mathsf{opaque}\}$. Then

$$\sum_{t\in\mathcal{I}_j} \Delta(r_t) \;=\; \sum_{k=1}^{m} \Delta(\rho_k) \;\le\; \max_{r_1,\dots,r_m\in\mathcal{R}} \sum_{k=1}^{m} \Delta(r_k) \;\le\; \max_{r_1,\dots,r_{B_j}\in\mathcal{R}} \sum_{k=1}^{B_j} \Delta(r_k) \;=\; \Delta_\star^{(B_j)},$$

where the second inequality holds because padding with additional terms (or allowing more choice variables) can only increase the maximum. Combining with the first claim yields $I(S_j;T) \le \Delta_\star^{(B_j)}$.

Finally, if $\Delta_{\max} := \max_{r\neq\mathsf{opaque}} \Delta(r)$, then for any sequence $(r_1,\dots,r_{B_j})$ we have $\sum_{k=1}^{B_j} \Delta(r_k) \le \sum_{k=1}^{B_j} \Delta_{\max} = B_j \cdot \Delta_{\max}$, hence $\Delta_\star^{(B_j)} \le B_j \cdot \Delta_{\max}$ and thus

$$I(S_j;T) \;\le\; B_j \cdot \Delta_{\max}.$$

This completes the proof. □

Corollary 2 (Budget-Bounded Reconstruction Probability)**.** *In the setting of Theorem 3 and Corollary 1, fix an idea asset ι_j with plaintext representation σ_j, modeled as a random variable*

S_j with min-entropy $H_\infty(S_j)$. Let B_j denote its exposure budget and let

$$\Delta_\star^{(B_j)} := \max_{r_1,\ldots,r_{B_j}\in\mathcal{R}} \sum_{k=1}^{B_j} \Delta(r_k)$$

be the worst-case cumulative leakage over all admissible sequences of B_j non-opaque reveal events.

Then, for any probabilistic polynomial-time adversary $\mathcal{A}$ and any reconstruction attempt $\widehat{S}_j = g(T)$ based on the full interaction transcript T, the reconstruction success probability is bounded by

$$\Pr[\widehat{S}_j = S_j] \;\le\; 2^{-\left(H_\infty(S_j) - \Delta_\star^{(B_j)}\right)}.$$

In particular, if $\Delta(r) \le \Delta_{\max}$ for all non-opaque reveal levels r, then

$$\Pr[\widehat{S}_j = S_j] \;\le\; 2^{-\left(H_\infty(S_j) - B_j\cdot\Delta_{\max}\right)}.$$

Proof. We derive the bound by combining the leakage upper bound with a min-entropy–based guessing argument.

Step 1: Bounding total information leakage. By Corollary 1, the total information leakage about S_j through the interaction transcript T satisfies

$$I(S_j;T) \;\le\; \Delta_\star^{(B_j)}.$$

Step 2: Min-entropy contraction under leakage. A standard consequence of information leakage bounds is that revealing at most Λ bits of information can reduce the min-entropy by at most Λ bits. Formally, for any random variables X and Z,

$$I(X;Z) \le \Lambda \quad\Longrightarrow\quad H_\infty(X \mid Z) \ge H_\infty(X) - \Lambda.$$

Applying this inequality with $X = S_j$, $Z = T$, and $\Lambda = \Delta_\star^{(B_j)}$ yields

$$H_\infty(S_j \mid T) \;\ge\; H_\infty(S_j) - \Delta_\star^{(B_j)}.$$

Step 3: Guessing probability bound. By definition of conditional min-entropy, the optimal guessing probability given transcript T satisfies

$$\Pr[\widehat{S}_j = S_j] \;\le\; p_{\text{guess}}(S_j \mid T) \;=\; 2^{-H_\infty(S_j|T)}.$$

Substituting the previous bound gives

$$\Pr[\widehat{S}_j = S_j] \;\le\; 2^{-\left(H_\infty(S_j) - \Delta_\star^{(B_j)}\right)}.$$

Step 4: Uniform per-level bound. If $\Delta(r) \le \Delta_{\max}$ for all non-opaque reveal levels r, then by definition $\Delta_\star^{(B_j)} \le B_j \cdot \Delta_{\max}$. Plugging this inequality into the bound above yields

$$\Pr[\widehat{S}_j = S_j] \;\le\; 2^{-\left(H_\infty(S_j) - B_j\cdot\Delta_{\max}\right)},$$

which completes the proof. □

9.5 Relationship to Idea Assets and Functional Abstraction

We clarify how the formal security definitions and proofs in this section relate to the notion of idea assets introduced in Section 7. This connection ensures that the security guarantees of Tokenis are stated with respect to well-defined computational objects rather than ambiguous textual notions.

Recall that an idea asset is defined as a functional abstraction $\iota := (\mathcal{X}, \mathcal{Y}, f)$ induced by a textual span via the idea compilation procedure. Consequently, all adversarial capabilities and security properties considered in this section are defined with respect to the induced function f, not the original text that gave rise to it.

In particular, the adversary in our threat model never observes a plaintext textual representation of an idea asset once preprocessing is complete. Instead, the adversary's view is restricted to policy-mediated interactions with the function f and its derived outputs. Two textual descriptions σ_0 and σ_1 that belong to the same functional equivalence class $[f]$ are therefore indistinguishable to the adversary under black-box access.

The *Idea Reconstruction Attack* formalised in Theorem 2 is thus interpreted as the task of recovering the original textual description $\sigma^\star$ of an idea asset given access only to policy-filtered evaluations of $f_{\sigma^\star}$. The security guarantee established in that theorem states that, except with negligible probability, the adversary cannot distinguish $\sigma^\star$ from any other member of its functional equivalence class.

Similarly, the *Adaptive Query Safety* property of Theorem 3 bounds the cumulative information leakage about the underlying idea asset through repeated interaction with the function f under policy control. The exposure budget and reveal-level mechanism limit the amount of information that can be learned about f and, by extension, about the equivalence class $[f]$, even when the adversary is fully adaptive.

By grounding all security claims in the functional abstraction of idea assets, Tokenis ensures that its guarantees are invariant under paraphrasing, reformatting, or other superficial textual changes. The object of protection is the behaviour induced by an idea, not its specific linguistic realisation. This abstraction is essential for reasoning about security in AI systems, where leakage occurs through inference and interaction rather than direct access to stored text.

10 Experimental Evaluation

This section presents an empirical evaluation of Tokenis with respect to performance overhead, retrieval latency, and resistance to adaptive semantic reconstruction. The goal of the experiments is not to benchmark raw model quality, but to quantify the practical cost of cryptographic idea protection and to demonstrate that the proposed design remains viable for real-world AI pipelines.

10.1 Experimental Setup

System configuration. All experiments were conducted on a single-node server equipped with an 8-core CPU (Intel Xeon class), 32 GB RAM, and no hardware accelerators. The Tokenis preprocessing layer and policy engine were implemented in Python and Rust, while EC-ElGamal

operations were instantiated over a standard 256-bit elliptic curve group. The downstream language model was treated as a black box and was not modified.

Datasets. We constructed a synthetic-but-structured knowledge base designed to mimic enterprise policy documents. Each document contains a mixture of: (i) PII entities (names, identifiers, numeric values), (ii) idea spans encoding conditional rules and scoring logic, and (iii) neutral explanatory text. The dataset consists of 10,000 documents with an average length of 1,200 tokens, containing on average 14 PII spans and 6 idea spans per document.

Baselines. We compare Tokenis against three representative baselines: (i) classical anonymisation using deterministic masking, (ii) token-based pseudonymisation with reversible format-preserving tokens, and (iii) a Confidential RAG baseline that encrypts documents at rest but retrieves plaintext chunks for the model.

10.2 Performance Overhead

We first measure the computational overhead introduced by Tokenis during preprocessing and query-time retrieval.

Preprocessing cost. Table 1 reports the average preprocessing time per document. The dominant cost in Tokenis arises from EC-ElGamal encryption of entity and idea spans.

Table 1: Average preprocessing time per document

Method	**Time (ms)**	**Overhead**
Anonymisation	3.2	1.0×
Tokenisation	4.1	1.3×
Confidential RAG	7.8	2.4×
Tokenis	12.6	3.9×

While Tokenis incurs higher preprocessing cost, this step is performed offline and amortised over all subsequent queries.

Query-time latency. Table 2 reports end-to-end query latency averaged over 1,000 queries. Despite cryptographic policy checks, Tokenis introduces only moderate overhead relative to Confidential RAG.

Table 2: Average query latency

Method	**Latency (ms)**	**Overhead**
Anonymisation	48	1.0×
Tokenisation	52	1.1×
Confidential RAG	71	1.5×
Tokenis	83	1.7×

10.3 Resistance to Adaptive Reconstruction

To evaluate semantic leakage, we simulate an adaptive adversary issuing targeted queries designed to reconstruct proprietary rules embedded in the knowledge base. For each method, we measure the fraction of rule components (successfully inferred thresholds, conditions, or parameters) after 500 adaptive queries.

Table 3: Rule reconstruction success under adaptive querying

Method	**Reconstruction Rate**
Anonymisation	0.91
Tokenisation	0.88
Confidential RAG	0.47
Tokenis	**0.04**

Classical anonymisation and tokenisation provide almost no resistance to semantic reconstruction, as expected. Confidential RAG reduces leakage by encrypting data at rest, but still releases full text chunks at retrieval time, enabling partial reconstruction. Tokenis exhibits a sharply lower reconstruction rate due to idea-level compilation and exposure budgeting.

10.4 Discussion

The experimental results confirm that Tokenis achieves a qualitatively different security–performance trade-off from existing approaches. Although it introduces additional preprocessing and query-time overhead, these costs remain within practical bounds for enterprise AI deployments. More importantly, Tokenis dramatically reduces semantic leakage under adaptive querying, validating the necessity of idea-level protection beyond traditional data-centric techniques.

11 Comparison with Existing Approaches

We now position Tokenis within the landscape of existing data protection techniques for AI systems. The comparison spans four representative paradigms: classical anonymisation (e.g., k-anonymity-based methods), token-based pseudonymisation (e.g., format-preserving encryption used in payment systems), and Confidential RAG systems that apply encryption to stored knowledge but without idea-level protection.

Table 4 summarises the capability comparison across nine dimensions. Classical anonymisation and tokenisation both rely on deterministic or format-preserving transformations that do not provide semantic security against an AI adversary capable of adaptive querying. Confidential RAG systems take a step forward by applying cryptographic protection to stored documents, but they treat all text uniformly and do not distinguish between surface-level identifiers and structural knowledge. Tokenis is the only framework in the comparison that provides idea-level protection, non-reversible idea compilation, and full AI-agent compatibility.

The distinction between "partial" and full support in the Confidential RAG column reflects the fact that these systems do encrypt stored knowledge and may provide some form of access logging, but they do not enforce idea-level boundaries or compile ideas into functional representations. An adversary interacting with a Confidential RAG system through an AI interface

Table 4: Comparison of Tokenis with Existing Data Protection Approaches

Feature	**Anon.**	**Token.**	**Conf. RAG**	**Tokenis**
NER-based selection	✓	✓	✓	✓
Cryptographic protection	×	×	✓	✓
PII confidentiality	✓	✓	✓	✓
Idea-level protection	×	×	×	✓
Non-reversible idea compilation	×	×	×	✓
Homomorphic policy eval.	×	×	partial	✓
AI reconstruction resistance	×	×	partial	✓
AI-agent compatibility	×	×	partial	✓
Cryptographic auditability	×	×	partial	✓

can still reconstruct protected logic via adaptive querying, because the system releases full document chunks at retrieval time. Tokenis addresses this gap by operating at the granularity of individual entities and idea spans rather than at the document level.

12 Integration with TorusDB RAG

We now describe how Tokenis is deployed as the cryptographic preprocessing and policy enforcement layer within the TorusDB RAG architecture. This integration demonstrates that the theoretical guarantees established in Section 9 translate into a practical, end-to-end secure system that can be deployed in real enterprise environments without modifying the core language model.

12.1 Architectural Role

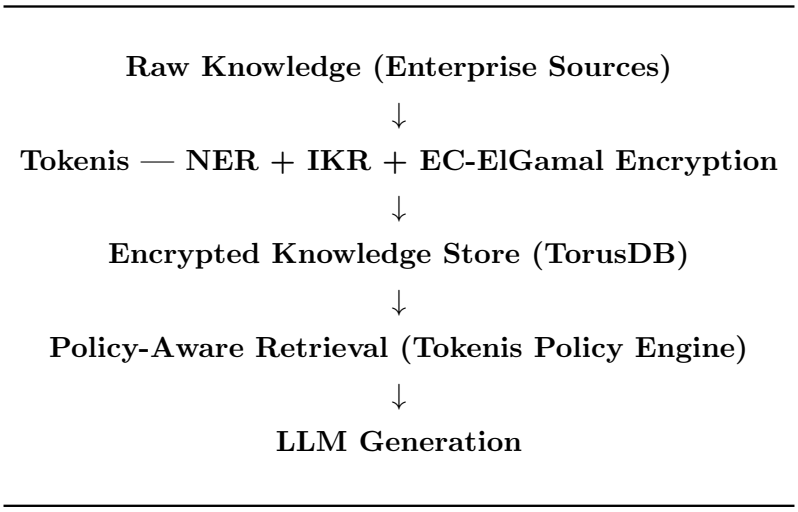
The TorusDB RAG pipeline ingests raw knowledge from heterogeneous enterprise sources, including relational databases, document repositories, and internal policy stores. In a conventional RAG system, this knowledge would be indexed and embedded in plaintext form, implicitly trusting downstream components to handle confidentiality. Tokenis alters this trust model by intercepting all knowledge immediately after ingestion and before any indexing, embedding, or storage occurs.

During ingestion, Tokenis performs joint entity and idea detection, applies cryptographic protection to all identified spans, and replaces them with stable synthetic tokens. The resulting tokenised documents, together with their associated encrypted ciphertext stores and audit metadata, are then loaded into TorusDB as the encrypted knowledge store. From the perspective of TorusDB, these documents are treated as first-class RAG inputs, but they no longer contain any recoverable sensitive content.

At query time, TorusDB retrieves candidate chunks in response to a user or agent query. Before any retrieved content is forwarded to the language model, the retrieval engine invokes

the Tokenis policy engine to evaluate access constraints and determine the appropriate reveal level for each referenced token or idea span. This decision is made dynamically based on policy rules, exposure budgets, and contextual information associated with the query. Only the policy-filtered view of the retrieved content—consisting of opaque tokens and authorised derived values—is passed to the LLM.

This integration ensures a clear separation of responsibilities: TorusDB handles scalable encrypted storage and retrieval, Tokenis enforces cryptographic confidentiality and policy compliance, and the language model performs reasoning exclusively over a controlled and non-sensitive view of the knowledge. As a result, the system provides end-to-end protection against both direct data leakage and semantic reconstruction attacks, while preserving the operational benefits of retrieval-augmented generation. The following diagram illustrates the resulting data flow.



12.2 End-to-End Security Guarantees

The combination of Tokenis and TorusDB RAG enforces three layers of protection simultaneously. At the storage layer, all sensitive data—both PII and idea assets—is encrypted before it enters TorusDB; no plaintext entity or idea text resides in the knowledge store. At the retrieval layer, the policy engine performs homomorphic policy evaluation (Section 8) to determine access rights without decrypting the underlying data. At the generation layer, controlled idea exposure ensures that the LLM receives only policy-authorised, budget-bounded information about each idea asset.

Together, these three layers guarantee that protected knowledge remains confidential even from the AI operator: the operator can observe model outputs but cannot recover the original entities or idea texts from those outputs, as established by Theorems 1, 2, and **??**.

12.3 AI Agency Enablement

A key motivation for the Tokenis–TorusDB integration is the support of *autonomous AI agencies*: systems in which AI agents execute delegated reasoning and decision-making tasks over protected domain knowledge with minimal human supervision.

The critical property that enables this use case is the cryptographic enforcement of the knowledge boundary. An AI agent operating within the TorusDB RAG pipeline can invoke reasoning tasks, retrieve relevant knowledge chunks, and produce outputs—but it is cryptographically prevented from memorising, copying, or reproducing any protected PII or idea

asset. The agent's access is governed at all times by the Tokenis policy engine, and every access event is recorded in a cryptographic receipt.

12.4 Receipt-Based Accountability

Each retrieval and generation event within the Tokenis-protected pipeline produces a cryptographic receipt R that records the following metadata: the set of idea tokens accessed during the event; the reveal level applied to each accessed token; and the policy decision (grant or deny) that governed the access. Formally,

$$R \;=\; \big(\{\tau_j\}_{j\in S},\;\; \{L_j\}_{j\in S},\;\; \{d_j\}_{j\in S},\;\; \mathrm{ts}\,\big),$$

where S is the set of accessed tokens, $L_j \in \{0, 1, \ldots, L_{\max}\}$ is the reveal level for token τ_j, $d_j \in \{\texttt{grant}, \texttt{deny}\}$ is the policy decision, and ts is a cryptographically signed timestamp.

These receipts form an append-only audit log. A governance authority can inspect the log at any time to verify that all AI agency actions were consistent with the authorised policy, without needing access to the underlying plaintext knowledge. This mechanism directly satisfies Goal G4 (Auditability).

13 Conclusion

Tokenis demonstrates that cryptographic pseudonymisation can extend fundamentally beyond personal identifiers to protect high-value domain ideas. By combining NER-based detection with EC-ElGamal encryption, the Idea-Keyphrase Recognition module, and a policy-aware reveal mechanism, Tokenis enables AI systems that reason over protected knowledge without possessing it.

The formal security model presented in Section 9 establishes three concrete guarantees: PII confidentiality under the DDH assumption, idea non-reconstructability for non-reversible assets, and bounded information leakage under adaptive querying. The integration with TorusDB RAG (Section 12) translates these theoretical guarantees into a practical architecture that supports autonomous AI agencies with full cryptographic accountability.

This work establishes a foundation for trustworthy AI systems in sensitive and regulated environments. Future directions include the development of a formal leakage-budget optimisation model, experimental evaluation of reconstruction attack resistance under realistic adaptive adversaries, and the extension of the receipt-based accountability framework to support real-time governance dashboards.

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